

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 50

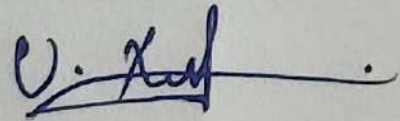
Annexure A

CONCEPT MAPPING

Code of Conduct:

I V. Yuvanshan Kar (192511265) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Security Attacks, Threats, Risks and Security Goals

Goals :

1) Concept map :

② Introduction

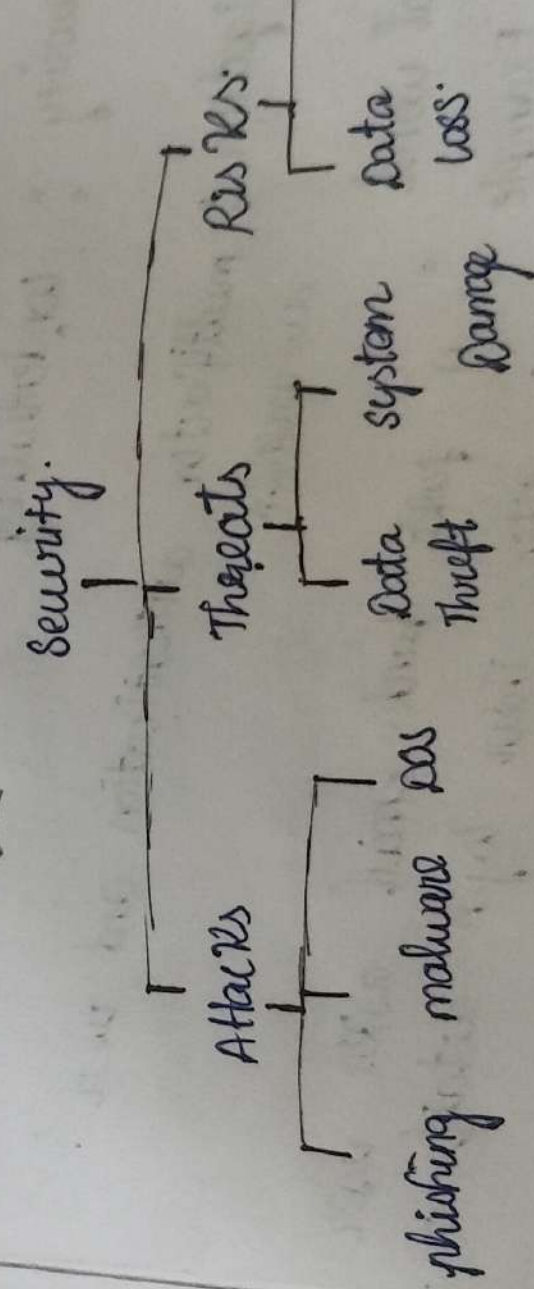
Information Security protects computers, networks and data from the unauthorized modification, destruction and interruption.

Attack is a intentional attempt to exploit weakness in system. An attack creates or threat which may result risk to

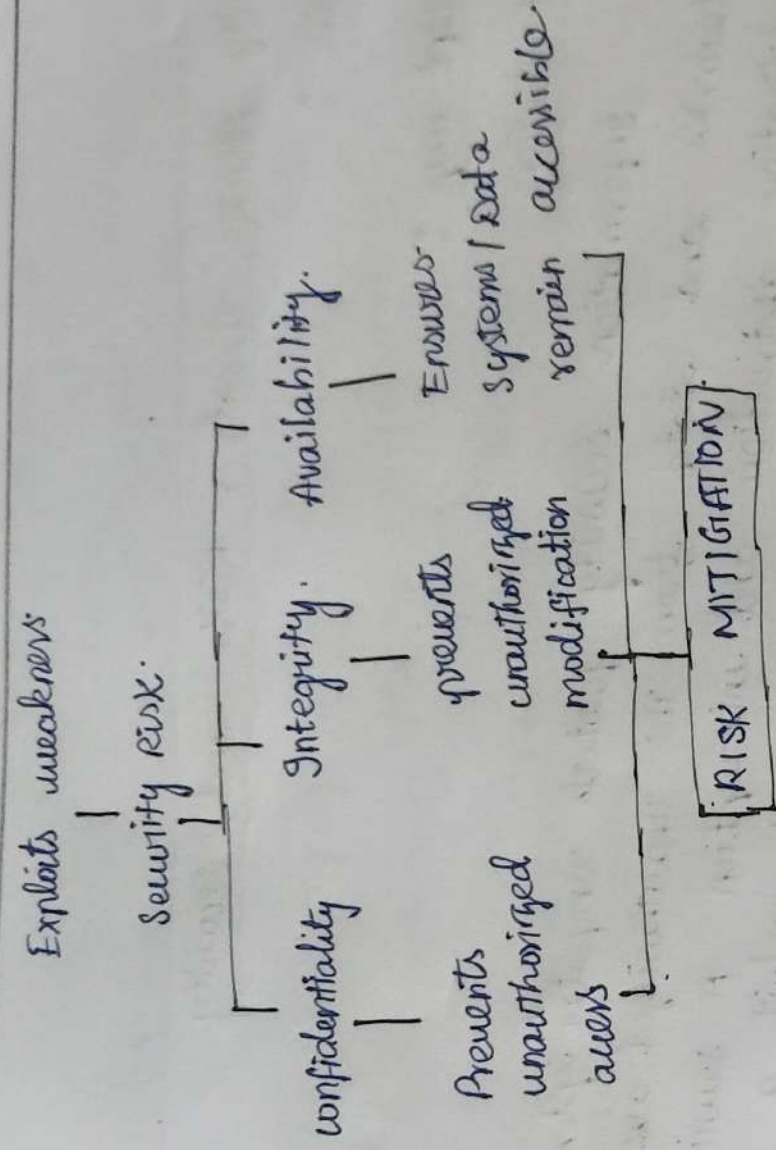
Security goals such as confidentiality, Integrity, Availability are used to reduce risk and

Information.

Concept map :



Phishing Attack



1) Phishing Attack :

① Attack : phishing uses fake emails to trick

② Threat : Theft of usernames, password.

③ Risk : unauthorized access and identification

④ Security : Confidentiality

⑤ Mitigation : multifactor authentication and user awareness

⑥ Real world

Example : A fake bank email asking them to login on their website.

2. MALWARE ATTACK :

- * Attack : malware is malicious software that can damage or access a system
- * Threat : Data may be stolen modified or destroyed
- * Risk : Data loss and system disruption
- * Security : Integrity and availability
- * Mitigation : Antivirus software, regular update and
- * Real world Example : Ransomware encrypts an organization's files and prevents user from accessing them.

3. Denial of Service Attack :

- ① Attack : The attacker sends a large number of requests to the server.
- ① Threat : The server becomes overloaded.
- ① Risk : Legitimate users cannot access the service.
- ① Security : Availability.
- ① Mitigation : Firewalls, traffic filtering and rate limiting.
- ① Real world Example : An online shopping website becomes unavailable because of a Denial of Service attack.

concept Map :

steganography

hiding

Security Goals - CIA :

Confidentiality : Ensures that information only by authorised users

Integrity : Ensures data is not changed modified by unauthorised users

Availability : Ensures that system and info available when required.

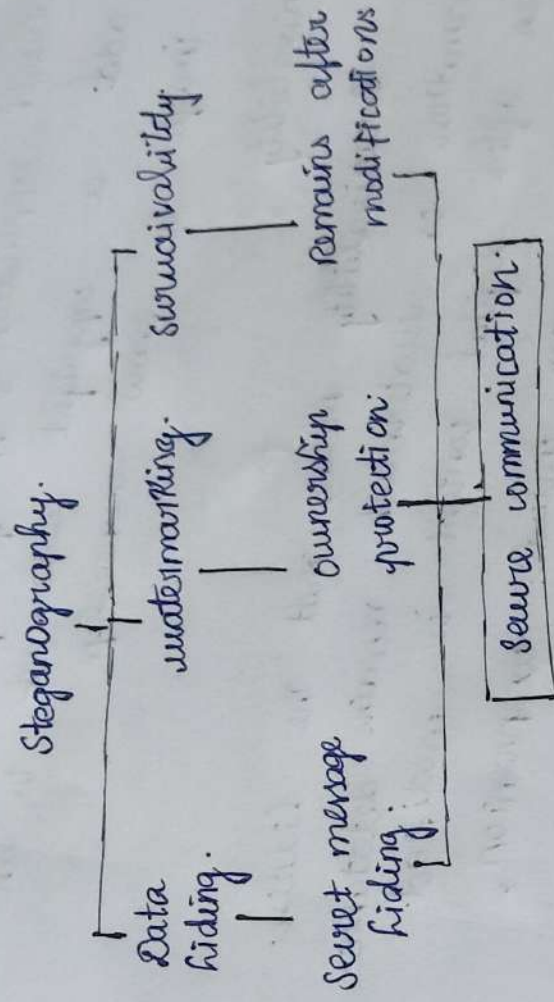
Overall Relationship :

Attack $\rightarrow$ Threat $\rightarrow$ Risks $\rightarrow$ CIA security
 $\downarrow$
Mitigation

Thus security attacks creates threats and the CIA security goals help organizations protect their information and reduce risks.

4) Steganography - Data hiding, water marking and. Survivability :

Concept Map :



1) Data hiding :

Data hiding is the process of hiding secret information inside another medium such as image, audio, video or document.

purpose : To conceal the existence of secret communication purpose.

Real world Explanation : A secret message can be hidden inside an image and sent to receiver.

2) watermarking :

watermarking is the process of embedding information such as ownership or copyright details into digital content.

Purpose : To protect ownership and authenticity of content.

Real world application : A photographer can add a copyright watermark to an image to show ownership.

3) Survivability :

Survivability means that the hidden information or watermark remains detectable even after digital content is modified. modifications may include compression, resizing or format conversion.

Purpose : To ensure that the embedded information remains useful after processing.

Real world application : A watermark in video should remain detectable even after the video is compressed for online sharing.

Relationship between concepts :

- ① Data hiding - hides secret information
- ② water marking - protects ownership and authenticity.
- ③ Survivability - keeps the hidden information effective after modification.

Together, these concepts support secure communication and protection of digital information.

Conclusion :

Steganography provides a way to hide information with in normal digital content. Data hiding provides security, ownership protection, and survivability that embedded information remains available after modifications.